

Day 6 - Interview Questions

1. What is the purpose of the while statement in Python?

The while statement is used for creating loops in Python. It allows you to repeatedly execute a block of code as long as a specified condition is true.

2. Explain the break and continue statements in the context of a while loop.

The break statement is used to exit a loop prematurely. When encountered, it immediately terminates the loop, even if the loop condition is still true.

The continue statement is used to skip the current iteration of a loop and proceed to the next iteration. It allows you to skip specific iterations without exiting the loop.

3. What is an "infinite loop," and how can you create one intentionally using a while statement?

An "infinite loop" is a loop that never terminates naturally because its condition always remains true. You can intentionally create an infinite loop using a while statement with a condition that is always true.

4. How can you add an "exit condition" to an infinite loop to break out of it?

To add an "exit condition" to an infinite loop, you can use an if statement within the loop to check a specific condition, and if that condition is met, you can use the break statement to break out of the loop.

5. Provide an example of an intentional infinite while loop in Python and explain a scenario where it might be used.

An example of an intentional infinite loop is a server program that continuously listens for incoming connections. It can be used in scenarios where the program needs to run indefinitely and perform a specific task continuously, such as a web server listening for incoming requests.

6. What is the purpose of the break statement in Python, and how does it work within loops?

The break statement is used to exit a loop prematurely. When encountered, it immediately terminates the loop, even if the loop condition is still true. It's commonly used to exit a loop based on a specific condition.

7. Explain the key differences between an if statement and a break statement in Python.

An if statement is used for conditional branching within the code, allowing different paths of execution based on a condition.

A break statement is used to exit a loop prematurely and does not affect the normal flow of the code outside the loop.

8. What is the purpose of the continue statement in Python, and how does it work within loops?

The continue statement is used to skip the current iteration of a loop and proceed to the next iteration. It allows you to skip specific iterations without exiting the loop entirely.

9. Explain a scenario where you might use the continue statement in a loop.

The continue statement can be used when you want to skip certain elements or items in a loop iteration, such as skipping over specific values in a list during processing.

10. What happens if the condition in a while loop is initially false?

If the condition in a while loop is initially false, the block of code inside the loop is never executed, and the loop terminates immediately.